

Finite-Core Universality and Sharp Large Order for Countable Exponential-Feedback Transseries

A sharp cutoff, a moving Gaussian saddle,
and superquadratic inverse asymptotics

Research article prepared for Vladimir Reshetnikov
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Abstract

For the formal equation

$$\widehat{U}(q) = \sum_{j \geq 1} q^j \exp(a j^p \widehat{U}(q)), \quad a > 0, \quad p > 1,$$

we pass from logarithmic large-order estimates to a full multiplicative coefficient equivalent. The main step is a sharp finite-core reduction. Keep actions $1, \dots, M$ in an analytic implicit core, and retain exactly one further action through its linear response. The coefficients of this one-tail series are asymptotic to those of $\widehat{U}$ if and only if $M(p-1) > 1$. At equality the ratio tends to zero, not one. A coupled small-radius/large-action saddle then gives the full exponential factor and determinant prefactor using only the finite core. It also yields a conditional local Gaussian law and an unconditional central limit theorem for the largest action, centered at an explicitly defined moving saddle, with standard deviation asymptotic to $\sqrt{pn}/((p-1)\log n)$.

The result allows nonnegative changes to finitely many initial slopes and weights, with the first weight kept equal to one. For $p > 2$ the finite core disappears from the leading multiplicative equivalent. For $p = 2$ a factor $\exp((1+1/a)r_n^2)$ survives, where $r_n(1+r_n)e^{2r_n} = an$; discarding it produces an unbounded relative error. For $p > 2$ we additionally prove a sharp formal-reversion law, $v_n = -u_n + (n+1)u_2u_{n-1} + o(nu_{n-1})$. The proofs are self-contained apart from elementary analytic and formal power-series tools. Exact coefficient checks, numerical saddle diagnostics, a proof-dependency audit, and a further research program accompany the text.

Status. This is an AI-assisted research draft with conventional mathematical proofs, not a Lean-verified development. The model-specific theorems below are the proposed contributions. Their global originality and publication priority have not been established. No named longstanding conjecture, general summability theorem, or full inverse theorem in the quadratic/subquadratic regime is claimed solved.

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1 The problem left by the repository

1.1 A precise extension, rather than another coefficient list

The ProveIt transseries project develops asymptotic scales, power-logarithmic and exponential expansions, formal composition and reversion, analytic realization, and quantitative inversion. Its current documentation carefully distinguishes the scope of the Lean modules from the human research manuscripts [1]. In particular, a formal expansion and a theorem about the growth or realization of its coefficients are different objects.

The recently filed article *A sharp regularity classification for countable exponential-feedback transseries* [2] studies

$$\widehat{U}(q) = \sum_{j \geq 1} q^j e^{\lambda_j \widehat{U}(q)}.$$

It obtains convergence and Gevrey classifications, logarithmic coefficient asymptotics, macroscopic concentration, and sectorial realizations. After its concentration proof it explicitly leaves a full multiplicative equivalent, a local limit theorem, and the fluctuation scale around a sufficiently accurate finite- n center unresolved. Its reversion result transfers Gevrey class membership, not sharp coefficient asymptotics.

Those are the gaps addressed here, for the substantial but specific class $\lambda_j = aj^p$, $p > 1$, and finite nonnegative perturbations of that class. The distinction in precision is important. Knowing $\log u_n = X_n + o(n)$ allows an undetermined factor $\exp(o(n))$. Such a factor can be larger than every power of n . We exhibit exactly this phenomenon at $p = 2$ and give a finite procedure that incorporates all factors needed for a relative error tending to zero.

The source inspected for [2] is pinned at repository commit 5804c7aff9955e6cbc09be2a71a295a07b238bc9; its TeX blob is c6a83d71973a12fe7ed970d2abc56e2b49e6808f. The relevant current directory is

```
Analysis/Transseries/docs/series-and-transseries/Exponential_Feedback_Regula
rity_Classification/.
```

The repository was changing during inspection. We do not claim an exhaustive audit of every incoming archive or every branch. Our main proofs do not assume an unreviewed theorem from an earlier research draft: the needed coefficient identity and macroscopic bounds are established again below.

1.2 Relation to established mathematics

Lagrange inversion is classical; Gessel’s survey [3] provides several formal and combinatorial forms. Infinite-colour inversion is also an established subject, with considerably broader settings than our countable scalar kernel [4]. Neither Lagrange inversion nor the passage to infinitely many formal colours is claimed as a new method here.

Condensation, finite clouds of small components, and threshold phenomena already occur in superexponentially weighted random trees. Janson, Jonsson, and Stefánsson [5] obtain detailed results for product degree weights, and Janson [6] surveys the relation to random allocations. Our coefficient weights instead contain $(\lambda_{j_1} + \cdots + \lambda_{j_m})^{m-1}$, not a fixed product of single-part weights. The growing action has scale $n/\log n$, and the small-action threshold carries logarithmic factors. Thus the familiar condensation picture is useful context, but is not itself our novelty claim or a substitute for the proofs below.

Rapid-coefficient expansions under formal operations have a classical antecedent in Bender’s work [7]. We give the specific convolution and reversion argument needed here, including its range

$p > 2$, rather than asserting that formal reversion automatically preserves a leading constant. For the formal transseries viewpoint, see also Edgar [8].

1.3 What the results do and do not provide

The main output is a multiplicative asymptotic theorem for a divergent one-generator transseries. Substituting $q = e^{-x}$ gives $\sum u_n e^{-nx}$, whose support is entirely elementary; the difficult information lies in the coefficient growth and the implicit feedback. No enlarged monomial support is required for the phenomenon.

The finite-core formula uses analytic functions only near the origin, where they are unambiguous. It never evaluates the original infinite kernel at a positive argument. This point is essential: for $p > 1$, that infinite sum diverges for every fixed positive pair (q, u) . Similarly, no Borel or Laplace integral is justified merely by obtaining precise large-order coefficients. Those analytic questions remain separate.

2 Model, finite core, and principal theorems

2.1 The admissible family

Fix $a > 0$, $p > 1$, and an integer $M_0 \geq 1$. Assume

$$w_1 = 1, \quad w_j, \lambda_j \geq 0, \quad w_j = 1, \quad \lambda_j = aj^p \quad (j > M_0). \quad (2.1)$$

The finitely many exceptional values are finite real numbers. Write

$$\widehat{U}(q) = \sum_{j \geq 1} w_j q^j e^{\lambda_j \widehat{U}(q)} = \sum_{n \geq 1} u_n q^n. \quad (2.2)$$

All infinite identities involving $\widehat{U}$ are formal. Coefficients of any fixed degree involve only finitely many summands. We prove below that the solution exists uniquely and $u_1 = 1$. The *pure model* has $w_j = 1$, $\lambda_j = aj^p$ for all $j \geq 1$.

Fix an integer $M \geq M_0$. Define the analytic core and response by

$$\Phi_M(t, u) = \sum_{j=1}^M w_j t^j e^{\lambda_j u}, \quad G_M(t) = \Phi_M(t, G_M(t)), \quad (2.3)$$

$$R_M(t) = (1 - \partial_u \Phi_M(t, G_M(t)))^{-1}. \quad (2.4)$$

The branches have $G_M(0) = 0$, $G'_M(0) = 1$, and $R_M(0) = 1$. They are analytic on a fixed complex neighborhood of zero. Define

$$W_M(q) = R_M(q) \sum_{j > M} q^j e^{aj^p G_M(q)}, \quad w_{n,M} = [q^n] W_M(q). \quad (2.5)$$

The notation $w_{n,M}$ is distinct from the primitive weights w_j . Although G_M and R_M are analytic, W_M is a *formal* series. Its coefficient $w_{n,M}$ counts exactly one primitive action above M ; it is not the coefficient of the truncation G_M itself.

Theorem 2.1 (Sharp finite-core cutoff). *Under (2.1),*

$$\lim_{n \rightarrow \infty} \frac{w_{n,M}}{u_n} = \begin{cases} 1, & M(p-1) > 1, \\ 0, & M(p-1) \leq 1. \end{cases} \quad (2.6)$$

For the pure model the smallest sufficient integer is

$$M_*(p) = \left\lfloor \frac{1}{p-1} \right\rfloor + 1. \quad (2.7)$$

All parameters, including M , are fixed as $n \rightarrow \infty$.

Editorial note (ProveIt, 2026-09-29). The boundary case $M(p-1) = 1$ is sharpened, consistently with (2.6), by the later package [10]. At $p = 1 + 1/M$ its Theorem `thm:boundary` gives $u_n \sim w_{n,M} \exp(r_n^{M+1}/a^M)$, an explicit missing factor that makes the ratio tend to zero as stated here; for $M = 1$, $p = 2$ that factor is $\exp(r_n^2/a)$, a part of the factor in (2.19). A finite limit $e^{-\lambda(\theta)}$ of the ratio, with a Poisson law for the first omitted actions, occurs there only in a window where the exponent $p = p_n$ approaches $1 + 1/M$ with n (Theorem `thm:window`), outside the fixed-parameter setting of this theorem.

2.2 A computable multiplicative equivalent

In the next theorem let $G = G_M$, $R = R_M$ to simplify notation. The saddle is a real pair (z_n, τ_n) , with z_n not required to be an integer. Set $k_n = n - z_n$ and $\Lambda_n = az_n^p$.

Theorem 2.2 (Finite-core saddle formula). *For every fixed $M \geq M_0$, there is, for all sufficiently large n , a unique solution in the asymptotic region specified below to*

$$\boxed{n = z_n + az_n^p \tau_n G'(\tau_n), \quad -\log \tau_n = apz_n^{p-1} G(\tau_n).} \quad (2.8)$$

It satisfies

$$z_n \sim \frac{p}{p-1} \frac{n}{\log n}, \quad \tau_n \sim \frac{1}{a} \left(\frac{p-1}{p} \right)^p \frac{(\log n)^p}{n^{p-1}}. \quad (2.9)$$

Define, with all core functions evaluated at τ_n ,

$$B_n = \Lambda_n(\tau_n G' + \tau_n^2 G''), \quad C_n = 1 + \frac{pk_n}{z_n}, \quad (2.10)$$

$$H_n = ap(p-1)z_n^{p-2}G, \quad \Delta_n = C_n^2 - H_n B_n. \quad (2.11)$$

Then $\Delta_n > 0$ eventually and

$$\boxed{w_{n,M} \sim \mathcal{A}_{n,M} := \frac{R_M(\tau_n)}{\sqrt{\Delta_n}} \exp(az_n^p G_M(\tau_n) - (n - z_n) \log \tau_n).} \quad (2.12)$$

Consequently $u_n \sim \mathcal{A}_{n,M}$ when $M(p-1) > 1$, whereas $\mathcal{A}_{n,M}/u_n \rightarrow 0$ when $M(p-1) \leq 1$.

The uniqueness assertion is local to, for example, $z \log n/n \in [c_-, c_+]$, where $0 < c_- < p/(p-1) < c_+$ are fixed and sufficiently close, and to the small branch of the core. It is not a claim about every possible positive solution of the displayed equations away from that region.

2.3 Fluctuations and inverse asymptotics

The positive Lagrange formula in Section 3 defines a probability distribution on ordered action configurations of total action n . Let J_n denote their largest action and let $\mathcal{E}_M$ be the event that exactly one action exceeds M .

Theorem 2.3 (Moving Gaussian law). *For every fixed $M \geq M_0$, put*

$$\sigma_n^2 = \frac{B_n}{\Delta_n} \sim \frac{z_n^2}{pn} \sim \frac{p}{(p-1)^2} \frac{n}{(\log n)^2}. \quad (2.13)$$

For each fixed $A < \infty$, uniformly over integers $|j - z_n| \leq A\sigma_n$,

$$\sigma_n \mathbb{P}(J_n = j \mid \mathcal{E}_M) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(j - z_n)^2}{2\sigma_n^2}\right) + o(1). \quad (2.14)$$

The conditional standardized variables converge in distribution to $\mathcal{N}(0, 1)$. If $M(p-1) > 1$, then also

$$\frac{J_n - z_n}{\sigma_n} \implies \mathcal{N}(0, 1) \quad \text{without conditioning.} \quad (2.15)$$

We deliberately distinguish the conditional *local* theorem from the unconditional *weak* theorem. A conditioning event of probability $1 - o(1)$ transfers weak convergence, but alone does not transfer errors on the smaller $1/\sigma_n$ scale of individual point probabilities.

For the pure model, let $r_n > 0$ be the unique solution of

$$r_n(1 + r_n)^{p-1} e^{pr_n} = an^{p-1}, \quad j_n^0 = \frac{n}{1 + r_n}, \quad k_n^0 = n - j_n^0, \quad (2.16)$$

and define

$$S_n = \frac{\exp(k_n^0(1 + pr_n))}{\sqrt{1 + 2pr_n + pr_n^2}}. \quad (2.17)$$

Theorem 2.4 (Explicit large order and superquadratic reversion). *For the pure model, if $p > 3/2$ then*

$$u_n \sim S_n \exp\left(\frac{(a+1)(k_n^0)^2}{a(j_n^0)^p}\right). \quad (2.18)$$

In particular, $u_n \sim S_n$ for $p > 2$, whereas for $p = 2$,

$$\boxed{u_n \sim S_n \exp((1 + 1/a)r_n^2)}. \quad (2.19)$$

If $p > 2$ and $\widehat{V} = \widehat{U}^{(-1)} = \sum v_n q^n$, then

$$v_n = -u_n + (n+1)u_2 u_{n-1} + o(nu_{n-1}), \quad v_n \sim -u_n \sim -S_n. \quad (2.20)$$

Moreover,

$$\frac{u_n}{u_{n-1}} \sim e^{pr_n} = \frac{an^{p-1}}{r_n(1 + r_n)^{p-1}}. \quad (2.21)$$

Editorial note (ProveIt, 2026-09-29). For $a = 1$ and slopes eventually equal to j^2 , (2.19) (with Proposition 10.1 for finitely many changed slopes) is also the main coefficient theorem, Theorem `thm:coefficient`, of the package [9], derived independently and without citing this article: the same root $r_n(1 + r_n)e^{2r_n} = n$, the same exponent and the same prefactor. That article adds slope tails $\alpha j + \eta j / \log j$, the rate $O((\log n)^2 / \sqrt{n})$, a total-variation Poisson law for the number of actions equal to two jointly with the largest action, and a positive-ray Borel equivalent.

3 Exact formal identities and their probability model

Lemma 3.1 (Formal existence and positive coefficient formula). *Equation (2.2) has a unique solution in $q\mathbb{R}[[q]]$, with $u_1 = 1$ and nonnegative coefficients. Its coefficients are*

$$u_n = \sum_{m=1}^n \frac{1}{m!} \sum_{\substack{j_1+\dots+j_m=n \\ j_i \geq 1}} \left(\prod_{i=1}^m w_{j_i} \right) \left(\sum_{i=1}^m \lambda_{j_i} \right)^{m-1}. \quad (3.1)$$

Equivalently, for finitely supported multiplicity vectors $\mathbf{m} = (m_1, m_2, \dots)$,

$$u_n = \sum_{\sum j m_j = n} \frac{\prod_j w_j^{m_j}}{\prod_j m_j!} \left(\sum_j \lambda_j m_j \right)^{\sum_j m_j - 1}. \quad (3.2)$$

For the one-part term the zeroth power is interpreted as 1, including when its base vanishes.

Proof. The coefficient of q^n on the right side of (2.2) uses only coefficients of $\widehat{U}$ of degrees at most $n-1$. This recursively defines a unique solution and proves nonnegativity and $u_1 = w_1 = 1$.

For the explicit expression, introduce a separate marker s and solve $U_s = s\Phi(q, U_s)$, where $\Phi(q, u) = \sum_{j \geq 1} w_j q^j e^{\lambda_j u}$. Classical formal Lagrange inversion gives

$$[s^m]U_s = \frac{1}{m} [u^{m-1}] \Phi(q, u)^m.$$

Expanding the product and the exponential gives (3.1) after setting $s = 1$. That substitution is valid coefficientwise: every factor of Φ has positive q -order, so only $m \leq n$ can contribute to $[q^n]$. Grouping ordered lists by their multiplicities gives (3.2). $\square$

For all sufficiently large n , the one-part configuration of size n has weight 1, so $u_n > 0$. Normalize every ordered summand in (3.1) by u_n . This defines a probability measure on the finite disjoint union of all compositions of n . Write $K_n = m$ for its number of parts, $J_n = \max_i j_i$, and

$$E_n = n - K_n + 1 = 1 + \sum_i (j_i - 1) \quad (3.3)$$

for one plus the total excess above unit actions. The integer E_n is not the largest part, although the two become asymptotically close.

Lemma 3.2 (The exact one-tail identity). *The core functions in (2.4) are analytic near zero with nonnegative Taylor coefficients. Furthermore,*

$$w_{n,M} = u_n \mathbb{P}(\mathcal{E}_M). \quad (3.4)$$

Proof. The analytic implicit function theorem applies to $u - \Phi_M(t, u)$ at $(0, 0)$, since its derivative in u is 1 there. Formal iteration proves nonnegative coefficients, and the analytic solution has the same Taylor series. The series $P(t) = \partial_u \Phi_M(t, G_M(t))$ has nonnegative coefficients and zero constant term; hence $R_M = (1 - P)^{-1}$ does too.

Mark every primitive action $j > M$ by s :

$$U_s = \Phi_M(q, U_s) + s \sum_{j > M} q^j e^{a_j U_s}.$$

At $s = 0$ the solution is G_M . Differentiating formally at $s = 0$ yields

$$(1 - \partial_u \Phi_M(q, G_M)) \partial_s U_s|_{s=0} = \sum_{j > M} q^j e^{a_j G_M}.$$

Thus $[s]U_s = W_M$. In (3.1), the power of s is exactly the number of parts above M , proving (3.4). $\square$

Example 3.3 (Small coefficients in the pure model). Directly from (3.2),

$$\begin{aligned} u_1 &= 1, & u_2 &= a + 1, \\ u_3 &= 1 + a(1 + 2^p) + \frac{3}{2}a^2, \\ u_4 &= 1 + a(1 + 2^p + 3^p) + \frac{1}{2}a^2(2 + 2^p)^2 + \frac{8}{3}a^3. \end{aligned}$$

For $a = 1, p = 2$, $u_3 = 15/2$ and the inverse coefficient is $v_3 = 2u_2^2 - u_3 = 1/2 > 0$. Thus an eventual-sign theorem for the inverse must not be misread as a sign assertion at every small degree.

4 Macroscopic localization, with full estimates

The following lemma establishes the coarse bounds needed later. The logarithmic conclusion is included to make the reduction self-contained; it is not presented as new relative to [2]. Put $s = p - 1$ and

$$X_n = sn \log n - pn \log \log n + \left(\log a + p \log \frac{p}{s} - s \right) n. \quad (4.1)$$

Lemma 4.1 (Exponential-scale localization). *Under (2.1), $\log u_n = X_n + o(n)$. For every $\delta > 0$ there is $c_\delta > 0$ such that*

$$\mathbb{P}\left(\left|\frac{E_n \log n}{n} - \frac{p}{p-1}\right| > \delta\right) \leq \exp(-c_\delta n + o(n)). \quad (4.2)$$

For every $\varepsilon > 0$ there is $d_\varepsilon > 0$ such that

$$\mathbb{P}(J_n < (1 - \varepsilon)E_n) \leq \exp(-d_\varepsilon n + o(n)). \quad (4.3)$$

Also, for every fixed $M \geq M_0$, $\log w_{n,M} = X_n + o(n)$, even if M is below the sharp cutoff.

Proof. Choose constants $D \geq 0$ and $W \geq 1$ so that $\lambda_j \leq aj^p + D$ and $w_j \leq W$ for all j . In a composition with m parts put $r = n - m + 1$. Convexity of x^p and repeated transfer of excess to a largest part give

$$\sum_i j_i^p \leq r^p + m - 1.$$

There are at most $r - 1$ nonunit parts, so their primitive weight product is at most W^{r-1} . Increasing D absorbs the extra $a(m - 1)$ and gives the row bound

$$[m\text{-part contribution}] \leq \binom{n-1}{r-1} W^{r-1} \frac{(ar^p + Dm)^{m-1}}{m!}. \quad (4.4)$$

Conversely, the configuration with one action r and $n - r$ unit actions contributes at least

$$T_{n,r} := \frac{(ar^p)^{n-r}}{(n-r)!}, \quad r > M_0. \quad (4.5)$$

The factor counting its possible positions cancels the difference between $m!$ and $(m - 1)!$.

Let $L = \log n$. Taking $r = \lfloor (p/s)n/L \rfloor$ in (4.5) and using Stirling's formula gives $\log T_{n,r} = X_n + o(n)$. We now show why no row improves this logarithm by order n . If $r \geq \eta n$ for fixed $\eta > 0$, the logarithm of the right side of (4.4) is at most $(1 - \eta)snL + O(n \log L)$, using the elementary entropy bound $\binom{n-1}{r-1} \leq 2^n$. Such rows are excluded from any maximizing sequence, so necessarily $r/n \rightarrow 0$.

In that range the binomial entropy and $(r-1)\log W$ are $o(n)$. Comparison with the lower bound forces $\log(ar^p + Dm) \geq pL - O(\log L)$; hence $\log r = L - O(\log L)$ and $Dm = o(r^p)$. Write $\alpha = r/n$. Uniform Stirling estimates now give, for any row within $O(n)$ of the maximum,

$$\frac{1}{n} \log[\text{row bound}] = (1 - \alpha)\{sL + p \log \alpha + \log a - \log(1 - \alpha) + 1\} + o(1).$$

Comparison once more with X_n/n gives $\alpha L = O(\log L)$. Consequently $\alpha \log \alpha = o(1)$ and, putting $y = rL/n$,

$$\frac{1}{n} \log[\text{row bound}] = sL - p \log L + \log a + 1 + p \log y - sy + o(1). \quad (4.6)$$

The function $g(y) = p \log y - sy$ tends to $-\infty$ at both ends of $(0, \infty)$ and is strictly concave, with its unique maximum at p/s . This proves the upper logarithmic estimate. It also proves uniform exponential loss away from any fixed neighborhood of p/s : otherwise a sequence of offending rows would, by the preceding localization, have a limit point contradicting the strict maximum of g . Summing at most n rows changes a logarithm by only $O(\log n)$. This proves (4.2).

For (4.3), it is enough to take $0 < \varepsilon < 1/2$. Restrict r to a fixed compact multiple of n/L , since its complement already has exponentially small weight. Under the constraint $j_i \leq b = \lfloor (1 - \varepsilon)r \rfloor$, convexity maximizes the slope sum by putting the excess into at most two nonunit parts, of sizes b and $r - b + 1$. Indeed, transferring mass between two interior nonunit parts increases the convex sum until one reaches 1 or the upper cap. Thus the slope sum is at most

$$a\{b^p + (r - b + 1)^p\} + D'm = ar^p\{(1 - \varepsilon)^p + \varepsilon^p + o(1)\}.$$

The expression in braces tends to a constant strictly below 1. Raising to the power $m - 1 \sim n$ loses a fixed exponential factor. The composition entropy and primitive weights are only $\exp(o(n))$ in this range and cannot compensate. The same maximum calculation proves (4.3).

Finally, $w_{n,M} \leq u_n$, while the lower configuration in (4.5) has exactly one part above any fixed M for large n . Therefore it is also a lower bound for $w_{n,M}$, proving its logarithmic asymptotic. $\square$

A useful consequence is that, with exponentially high probability, $K_n \sim n$, $E_n \asymp n/\log n$, and the largest action is unique. This still does *not* imply that all remaining actions are bounded by a particular M . That stronger assertion is the next step.

5 Why a sufficient finite core captures almost all weight

Proposition 5.1 (Sufficiency of the strict cutoff). *If $M(p-1) > 1$, then $\mathbb{P}(\mathcal{E}_M) \rightarrow 1$.*

Proof. Put $c_* = p/(p-1)$. Fix a small width $\delta > 0$ and a small $\varepsilon \in (0, 1/3)$, to be chosen below. By Lemma 4.1, except on a set of probability $\exp(-cn + o(n))$, a configuration has

$$(c_* - \delta) \frac{n}{\log n} \leq r = E_n \leq (c_* + \delta) \frac{n}{\log n}, \quad L_0 = J_n \geq (1 - \varepsilon)r. \quad (5.1)$$

The largest part L_0 is unique for large n , and is above M . Every other part $i > 1$ has $i - 1 \leq \varepsilon r$.

Consider a good configuration with an additional part $i > M$. Choose one such part by a fixed deterministic rule, for example its earliest position. Replace the pair (L_0, i) by $(L_0 + i - 1, 1)$, keeping all positions and the number m of parts unchanged. Both nonunit parts in this operation are in the pure tail, while the new unit part has weight $w_1 = 1$. Thus the primitive weight product and the denominator $m!$ in (3.1) are unchanged.

Let S and S' be the old and new slope sums. For every $\eta > 0$, choosing ε small enough gives, uniformly in (5.1) and in $i > M$,

$$S' - S \geq a(p - \eta)r^{p-1}(i - 1). \quad (5.2)$$

Indeed, the mean-value theorem bounds the gain at the large part below by $ap(1-\varepsilon)^{p-1}r^{p-1}(i-1)$. The loss is $ai^p - \lambda_1$. Since λ_1 is fixed and nonnegative, its possible contribution can be discarded, and $i^p/(i-1) \leq C_p(\varepsilon r + 1)^{p-1}$ for $i \geq 2$. Making ε small absorbs this term and proves (5.2) for large n .

By the convex bound used earlier, $S' \leq ar^p + Dm = ar^p(1 + o(1))$ uniformly here. The old coefficient weight divided by the new one is therefore at most

$$\left(\frac{S}{S'}\right)^{m-1} \leq \exp\left(-\frac{(p-2\eta)(m-1)(i-1)}{r}\right) \quad (5.3)$$

for sufficiently large n .

We record the multiplicity of this map, rather than treating it as an unjustified injection. In the image, the unique giant is still recognizable. Given its image configuration, the chosen slot, and i , the original configuration is recovered by subtracting $i-1$ from the giant and replacing the unit in that slot by i . Thus there are at most m preimages for each integer i . Let $\mathcal{B}_n$ denote the good configurations with an additional part above M . Summing (5.3) gives

$$\begin{aligned} \mathbb{P}(\mathcal{B}_n) &\leq \sup_{r \text{ in (5.1)}} n \sum_{i=M+1}^{\infty} \exp\left(-\frac{(p-2\eta)(n-r)(i-1)}{r}\right) \\ &\leq n^{1-\frac{(p-2\eta)M}{c_*+\delta}+o(1)}. \end{aligned} \quad (5.4)$$

The denominator of the geometric series tends to 1 and does not change the power of n . Because $M(p-1) > 1$, we may choose η and δ small enough that the exponent in the last line is negative. The exceptional configurations have exponentially small probability. On the good event at least one part exceeds M , so the absence of a second one is precisely $\mathcal{E}_M$. This proves the proposition. $\square$

Remark 5.2 (What makes this stronger than macroscopic condensation). The large-action result only says that the remaining excess is $o(n/\log n)$. The mass-transfer estimate controls every extra action above a *fixed* cutoff, and pays explicitly for its possible locations. The competition between a factor of order n and a suppression of order $n^{-M(p-1)}$ is the origin of the cutoff. It also explains why an unqualified statement that “one large action dominates” does not establish a multiplicative coefficient formula.

6 A uniform small-radius coefficient saddle

We next prove the analytic estimate used to compute $w_{n,M}$. Only a fixed analytic core is involved. In particular, no exchange of an infinite divergent tail with a contour integral will be made.

Lemma 6.1 (Coefficient estimate with a shrinking saddle). *Let $G(t) = t + O(t^2)$ and $R(t) = 1 + O(t)$ be analytic at zero with nonnegative Taylor coefficients. Fix $a > 0, p > 1$, and constants $0 < c_- < c_+ < \infty$. For integers*

$$c_- \frac{n}{\log n} \leq j \leq c_+ \frac{n}{\log n}, \quad k = n - j, \quad \Lambda = aj^p,$$

let $t = t_j > 0$ solve

$$\Lambda t G'(t) = k. \quad (6.1)$$

Then this solution exists uniquely on the small positive branch for large n , $t \sim k/\Lambda \rightarrow 0$ uniformly, and

$$[q^k]R(q)e^{\Lambda G(q)} = \frac{R(t)e^{\Lambda G(t)}t^{-k}}{\sqrt{2\pi B_j}}(1 + o(1)), \quad B_j = \Lambda(tG'(t) + t^2G''(t)). \quad (6.2)$$

The $o(1)$ is uniform over the indicated integers j .

Proof. Since $tG'(t) = t + O(t^2)$ has positive derivative near zero, the small solution exists uniquely whenever k/Λ is sufficiently small. In the stated range, $k/\Lambda \asymp n^{1-p}(\log n)^p \rightarrow 0$ uniformly. Also $B_j \sim k \sim n$.

Cauchy's formula on $q = te^{i\theta}$ gives

$$[q^k]R(q)e^{\Lambda G(q)} = \frac{e^{\Lambda G(t)}t^{-k}}{2\pi} \int_{-\pi}^{\pi} R(te^{i\theta}) \exp(\Lambda[G(te^{i\theta}) - G(t)] - ik\theta) d\theta.$$

The linear term vanishes by (6.1). With $D = t d/dt$, analyticity and $G(t) = t + O(t^2)$ give $\Lambda D^h G(t) = O(k)$ for every fixed positive integer h , uniformly as $t \rightarrow 0$. Hence, for $|\theta| \leq k^{-2/5}$,

$$\Lambda[G(te^{i\theta}) - G(t)] - ik\theta = -\frac{1}{2}B_j\theta^2 + O(k|\theta|^3),$$

where the error is $o(1)$ throughout this arc. The response satisfies $R(te^{i\theta})/R(t) = 1 + o(1)$ there.

For the complement, write $G(q) = \sum_{h \geq 1} g_h q^h$, with $g_1 = 1$ and $g_h \geq 0$. Then

$$G(t) - \Re G(te^{i\theta}) = \sum_{h \geq 1} g_h t^h (1 - \cos h\theta) \geq t(1 - \cos \theta). \quad (6.3)$$

Also $|R(te^{i\theta})| \leq R(t)$. Since $\Lambda t \asymp k$, the complementary integral is bounded by the main positive scale times $O(e^{-ck^{1/5}})$, using $1 - \cos \theta \geq c\theta^2$ on $[-\pi, \pi]$. The central Gaussian integral is $\sqrt{2\pi/B_j}(1 + o(1))$. This proves (6.2) and its uniformity. $\square$

The same proof allows a fixed analytic multiplier $t^d h(t)$ with $h(0) > 0$: its coefficient contribution is asymptotic to its value at the saddle times the right side of (6.2). The factor $e^{id\theta}$ is $1 + o(1)$ on the central arc. We will use this observation with fixed $d = M + 1$ in the sharpness proof.

7 The moving action saddle and its determinant

7.1 The exponent after coefficient extraction

For real j in a fixed compact multiple of $n/\log n$, define t_j by (6.1) and set

$$f_n(j) = aj^p G(t_j) - (n - j) \log t_j. \quad (7.1)$$

Here and throughout this section $G = G_M$, $R = R_M$. Differentiating the defining equation for t_j gives

$$\frac{t'_j}{t_j} = -\frac{C_j}{B_j}, \quad C_j = 1 + \frac{p(n-j)}{j}. \quad (7.2)$$

The saddle condition eliminates the terms containing t'_j when f_n is differentiated once, so

$$f'_n(j) = apj^{p-1}G(t_j) + \log t_j, \quad (7.3)$$

$$f''_n(j) = ap(p-1)j^{p-2}G(t_j) - \frac{C_j^2}{B_j}. \quad (7.4)$$

These are exact identities, not leading-order approximations.

Lemma 7.1 (Location, curvature, and uniqueness). *Choose fixed $c_- < p/(p-1) < c_+$. Uniformly on the interval $I_n = [c_- n/\log n, c_+ n/\log n]$,*

$$\frac{f'_n(cn/\log n)}{\log n} = \frac{p}{c} - (p-1) + o(1), \quad (7.5)$$

$$f''_n(j) = -\frac{pn}{j^2}(1 + o(1)), \quad f'''_n(j) = O\left(\frac{(\log n)^3}{n^2}\right). \quad (7.6)$$

For large n there is therefore exactly one zero z_n of f'_n in I_n , and it is the strict maximum of f_n there. The pair (z_n, t_{z_n}) is the solution asserted in Theorem 2.2.

Proof. In this interval $k = n - j \sim n$, $G(t) = t(1 + O(t))$, $G'(t) = 1 + O(t)$, and $t \sim k/(aj^p)$. Thus

$$apj^{p-1}G(t_j) = \frac{pk}{j}(1 + O(t_j)),$$

while

$$\log t_j = (1 - p) \log n + p \log \log n - \log a - p \log c + o(1)$$

when $j = cn/\log n$. This proves the first estimate and the signs at the two endpoints. In (7.4),

$$B_j = k(1 + O(t_j)), \quad ap(p-1)j^{p-2}G(t_j) = \frac{p(p-1)k}{j^2}(1 + O(t_j)).$$

Since $C_j = 1 + pk/j$, their difference is $-pk/j^2 - 2p/j - 1/k + o(n/j^2)$, proving the curvature estimate. Differentiating these analytic expressions, using $t'_j/t_j = O(1/j)$ and the corresponding derivatives of $G(t) = t + O(t^2)$, gives $f''_n = O(n/j^3)$, the stated bound. Strict negative curvature and the endpoint signs prove uniqueness. The leading location and the leading value of t_{z_n} follow immediately, giving (2.9). $\square$

7.2 Completing the two-stage asymptotic

Proof of Theorem 2.2. First localize the sum in (2.5) to $j \in I_n$, choosing c_-, c_+ on opposite sides of $p/(p-1)$. Under the one-tail event, j is the largest part. Lemma 4.1 shows that the contribution outside such an interval is at most $u_n \exp(-cn + o(n))$. Both u_n and $w_{n,M}$ have logarithm $X_n + o(n)$, so this is also exponentially negligible relative to $w_{n,M}$. This argument does not require the sufficient-cutoff result.

Inside I_n , Lemma 6.1 gives

$$w_{n,M} \sim \sum_{j \in I_n \cap \mathbb{N}} \frac{R(t_j)}{\sqrt{2\pi B_j}} e^{f_n(j)}. \quad (7.7)$$

At z_n , the identities (7.4) give

$$-f''_n(z_n) = \frac{\Delta_n}{B_n} > 0.$$

The estimates above yield

$$B_n \sim n, \quad \Delta_n \sim \frac{pn^2}{z_n^2}, \quad \sigma_n = \sqrt{\frac{B_n}{\Delta_n}} \rightarrow \infty. \quad (7.8)$$

For $h = j - z_n$ with $|h| \leq \sigma_n \log n$, Taylor's formula gives

$$f_n(z_n + h) = f_n(z_n) - \frac{h^2}{2\sigma_n^2} + o(1) \quad (7.9)$$

uniformly: the cubic error is $O((\log n)^3 n^{-2} (\sigma_n \log n)^3) = O((\log n)^3 / \sqrt{n}) = o(1)$. In the same window $R(t_j)/\sqrt{B_j}$ varies by a relative $o(1)$, since its logarithmic derivative is $O(1/j)$.

The curvature bound in (7.6) suppresses the remaining terms of I_n by $\exp(-c(\log n)^2)$ after the central window is removed. A polynomial number of terms cannot repair this loss. Finally, since $\sigma_n \rightarrow \infty$, the Gaussian lattice sum is a Riemann sum:

$$\sum_{j \in \mathbb{Z}} e^{-(j-z_n)^2/(2\sigma_n^2)} \sim \sqrt{2\pi} \sigma_n.$$

Multiplying this with the prefactor in (7.7) gives

$$w_{n,M} \sim R(\tau_n) e^{f_n(z_n)} \frac{\sigma_n}{\sqrt{B_n}} = \frac{R(\tau_n) e^{f_n(z_n)}}{\sqrt{\Delta_n}},$$

which is (2.12). The positive case of Theorem 2.1, already proved, transfers the formula to u_n when $M(p-1) > 1$. The negative case will follow from the next section. $\square$

Remark 7.2 (Why there is no missing factor of 2π). The coefficient saddle supplies $(2\pi B_n)^{-1/2}$. Summing the action saddle supplies $(2\pi)^{1/2} \sigma_n$. Their product is exactly $\Delta_n^{-1/2}$. Using only one saddle would miss both this cancellation and the moving action scale.

8 Sharpness: the boundary core is not sufficient

Proposition 8.1 (A single omitted small action forces failure). *For any fixed $M \geq M_0$, let $d = M + 1$. There is a nonnegative subseries of (2.2), disjoint from the one-tail subseries, whose coefficient $b_{n,M}$ satisfies*

$$\frac{b_{n,M}}{w_{n,M}} \sim k_n \tau_n^M. \quad (8.1)$$

In particular, if $M(p-1) \leq 1$, then $w_{n,M}/u_n \rightarrow 0$.

Proof. Temporarily add the action d to the core with a marker s :

$$G_s(q) = \Phi_M(q, G_s(q)) + s q^d e^{ad^p G_s(q)}. \quad (8.2)$$

Set

$$R_s(q) = (1 - \partial_u \Phi_M(q, G_s(q)) - s a d^p q^d e^{ad^p G_s(q)})^{-1}.$$

At $s = 0$, $G_s = G$ and $R_s = R$. Differentiation gives

$$D(q) := \partial_s G_s(q)|_{s=0} = R(q) q^d e^{ad^p G(q)} = q^d (1 + O(q)), \quad (8.3)$$

$$\dot{R}(q) := \partial_s R_s(q)|_{s=0} = R(q)^2 \left(\partial_u^2 \Phi_M(q, G(q)) D(q) + a d^p q^d e^{ad^p G(q)} \right). \quad (8.4)$$

Both series have nonnegative coefficients, and $\dot{R}(q) = O(q^d)$.

Now retain exactly one action $j > d$, and differentiate its response with respect to s at zero. The result is the formal nonnegative series

$$B_M(q) = \sum_{j>d} q^j e^{aj^p G(q)} \left(\dot{R}(q) + a j^p R(q) D(q) \right). \quad (8.5)$$

By the two independent markers, it counts configurations with exactly one part $j > d$, exactly one part d , and all remaining parts at most M . Consequently it is disjoint from W_M , and $u_n \geq w_{n,M} + b_{n,M}$ for $b_{n,M} = [q^n] B_M$.

For each j in the saddle region, apply Lemma 6.1 and its analytic-multiplier extension. Relative to the coefficient of $q^j R(q) e^{aj^p G(q)}$, the second summand in the parentheses in (8.5) contributes asymptotically $aj^p D(t_j) \sim aj^p t_j^d$. The first contributes $O(t_j^d)$ and is negligible, since $aj^p \rightarrow \infty$. The multiplier $aj^p t_j^d$ varies by a relative $o(1)$ across the Gaussian action window. Coarse tails are negligible by Lemma 4.1, since B_M is a subseries of $\hat{U}$ and $\log w_{n,M} = X_n + o(n)$. Inside the coarse interval but outside the Gaussian window, the multiplier has at most a polynomial effect, which cannot overcome the saddle decay. The finitely many excluded values $j \leq d$ have merely exponential coefficient growth and are negligible. Repeating the outer-saddle summation thus gives

$$\frac{b_{n,M}}{w_{n,M}} \sim a z_n^p \tau_n^d = \frac{k_n \tau_n^{d-1}}{G'(\tau_n)} \sim k_n \tau_n^M.$$

From (2.9), with $c_q = a^{-1}((p-1)/p)^p > 0$,

$$k_n \tau_n^M \sim c_q^M n^{1-M(p-1)} (\log n)^{pM}. \quad (8.6)$$

This diverges when $M(p-1) < 1$, and also at equality, where the positive power of $\log n$ remains. Since $u_n/w_{n,M} \geq 1 + b_{n,M}/w_{n,M}$, the claimed failure follows. $\square$

Proposition 8.1 completes Theorem 2.1 and the negative-cutoff conclusion of Theorem 2.2. It also identifies exactly what is wrong with replacing the strict inequality in (2.7) by a non-strict one. The boundary does not have a finite nonzero missing-action correction; that correction is already unbounded.

9 Gaussian fluctuations around the correct moving center

Proof of Theorem 2.3. On $\mathcal{E}_M$, the unique action $j > M$ is the largest. Its exact probability is

$$\mathbb{P}(J_n = j \mid \mathcal{E}_M) = \frac{[q^{n-j}]R_M(q)e^{aj^p G_M(q)}}{w_{n,M}}, \quad j > M. \quad (9.1)$$

Use (6.2) in the numerator and (2.12) in the denominator. For $|j - z_n| \leq A\sigma_n$, the ratio of the analytic prefactors tends to 1, while (7.9) gives the Gaussian exponent. The remaining constant is $\sqrt{\Delta_n}/\sqrt{2\pi B_n} = 1/(\sqrt{2\pi}\sigma_n)$, proving (2.14).

The same estimates on windows of growing standardized width, together with the curvature tail bound, imply conditional tightness and convergence of interval probabilities. This proves the conditional central limit theorem. If $M(p-1) > 1$, then $\mathbb{P}(\mathcal{E}_M) \rightarrow 1$ by the cutoff theorem. For every bounded continuous h , conditioning changes $\mathbb{E}h((J_n - z_n)/\sigma_n)$ by at most $2\|h\|_\infty \mathbb{P}(\mathcal{E}_M^c)$, which tends to zero. Thus the unconditional weak limit is the same. The variance scale in (2.13) follows from (7.8). $\square$

9.1 Why the leading action scale is not an adequate center

The approximation $z_n \sim (p/(p-1))n/\log n$ determines only the first scale. Even the next deterministic correction is generally much larger than $\sigma_n \asymp \sqrt{n}/\log n$. For the bare equation (2.16), for example,

$$pr_n = (p-1)\log n - p\log \log n + O(1),$$

so replacing r_n by its first term shifts j_n^0 by order $n \log \log n / (\log n)^2$. Dividing that shift by σ_n tends to infinity. A central limit theorem written with only the leading scale as its center would therefore usually be false.

The coupled equations (2.8) avoid this problem: they define a center accurate to the Gaussian scale without requiring a manually expanded series of logarithms. They are not an assertion that z_n equals the finite- n mean exactly. The theorem is the centered distributional limit, with the separately stated conditional local refinement.

10 Explicit equivalents and the quadratic correction

10.1 The bare saddle

Consider first the comparison sum

$$T_n = \sum_{j=1}^n \frac{(aj^p)^{n-j}}{(n-j)!}. \quad (10.1)$$

It corresponds to the same coefficient/action calculation with $G(t) = t$ and $R(t) = 1$. The saddle equations reduce to

$$k = aj^p t, \quad -\log t = \frac{pk}{j}, \quad j + k = n.$$

Writing $r = k/j$ gives exactly (2.16). Its left side is strictly increasing from 0 to ∞ , so the positive solution is unique. At this saddle,

$$t = e^{-pr}, \quad f = k(1 + pr), \quad B = k, \quad \Delta = (1 + pr)^2 - p(p-1)r^2 = 1 + 2pr + pr^2.$$

The preceding saddle proof therefore gives $T_n \sim S_n$, with S_n as in (2.17).

This comparison has a simple interpretation: the slope at the giant is kept, while every finite-core feedback correction is suppressed. Whether that suppression is harmless depends on p .

10.2 A controlled perturbation of the saddle

Proposition 10.1 (First core correction). *Under (2.1), write $\beta = u_2 = \lambda_1 + w_2$. If $p > 3/2$, then*

$$u_n \sim S_n \exp\left(\frac{\beta(k_n^0)^2}{a(j_n^0)^p}\right). \quad (10.2)$$

In particular, for $p > 2$ any finite perturbation satisfying (2.1) has $u_n \sim S_n$.

Proof. Choose a fixed $M \geq \max(M_0, 2)$ with $M(p-1) > 1$. Then $G_M(t) = t + \beta t^2 + O(t^3)$ and $R_M(t) = 1 + O(t)$. For a real j in the coarse saddle interval put $k = n - j$, $\Lambda = aj^p$, and $t_0 = k/\Lambda$. The small solution of $\Lambda t G'(t) = k$ has the uniform analytic expansion

$$t = t_0 - 2\beta t_0^2 + O(t_0^3).$$

Substitution into the exponent yields

$$\begin{aligned} f_G(j) &= k - k \log t_0 + \beta k t_0 + O(k t_0^2) \\ &= f_0(j) + \delta(j) + O(n t_0^2), \quad \delta(j) = \frac{\beta(n-j)^2}{a j^p}. \end{aligned} \quad (10.3)$$

The analytic remainder can be differentiated with respect to j ; since $t'_0/t_0 = -1/k - p/j = O(1/j)$, its first derivative is $O(n t_0^2/j)$. Similarly $\delta'(j) = O(n t_0/j)$.

Let $j_0 = j_n^0$ maximize f_0 and let z maximize f_G . Both lie in the same coarse interval, and its curvature has magnitude comparable to n/j_0^2 . The derivative estimates and the mean-value theorem therefore give $z - j_0 = O(j_0 t_0(j_0))$. Taylor's theorem then gives

$$f_G(z) - f_0(j_0) = \delta(j_0) + O(n t_0(j_0)^2). \quad (10.4)$$

Here the quadratic shift in the maximum, the change in δ , and the explicit remainder in (10.3) all have the displayed order. The determinant prefactors have ratio $1 + O(t_0(j_0))$, and $R_M(\tau_n) = 1 + o(1)$. These estimates follow directly from (2.11), because $z/j_0 = 1 + O(t_0)$ and the core and its first two derivatives differ from those of t by $O(t)$ in relative scale where appropriate.

Finally,

$$t_0(j_0) \asymp n^{1-p}(\log n)^p, \quad n t_0(j_0)^2 \asymp n^{3-2p}(\log n)^{2p} = o(1)$$

when $p > 3/2$. Combine (10.4) with Theorem 2.2 and the cutoff theorem. This proves (10.2). For $p > 2$, the retained correction $\beta k_0 t_0 = O(n^{2-p}(\log n)^p)$ also tends to zero. $\square$

Corollary 10.2 (Quadratic feedback). *For the pure quadratic model,*

$$r_n(1 + r_n)e^{2r_n} = an, \quad u_n \sim \frac{\exp\left(\frac{nr_n}{1+r_n}(1 + 2r_n) + (1 + 1/a)r_n^2\right)}{\sqrt{1 + 4r_n + 2r_n^2}}. \quad (10.5)$$

The ratio u_n/S_n tends to infinity faster than every fixed power of n .

Proof. Here $\beta = a + 1$ and $k_0/j_0 = r_n$, so the correction in (10.2) is $(1 + 1/a)r_n^2$. Also $r_n \sim \frac{1}{2} \log n$. Thus $\log(u_n/S_n) \sim (1 + 1/a)(\log n)^2/4$, which dominates every fixed multiple of $\log n$. $\square$

The order- n logarithmic asymptotic cannot detect this factor because $(\log n)^2 = o(n)$. The quadratic example is therefore a concrete reason to distinguish logarithmic control from multiplicative control.

10.3 The hierarchy below the quadratic regime

For $3/2 < p < 2$, the first correction in (10.2) grows as a positive constant times $n^{2-p}(\log n)^p$ in the pure model. The same two-core calculation still gives a relative equivalent, because the next exponent scale tends to zero. At $p = 3/2$, that next scale is logarithmically divergent; a two-action core is no longer enough. More generally, the small-action scales are

$$n\tau_n^{d-1} \asymp n^{1-(d-1)(p-1)}(\log n)^{p(d-1)}. \quad (10.6)$$

Theorem 2.1 handles the whole hierarchy by retaining an exact finite analytic core instead of expanding each of these exponent scales by hand. In the pure model, representative choices are

Range of p	Smallest sufficient M
$p > 2$	1
$3/2 < p \leq 2$	2
$4/3 < p \leq 3/2$	3
$5/4 < p \leq 4/3$	4

The core must grow as $p \downarrow 1$. None of the fixed- p arguments above asserts a uniform theorem in that limit.

11 Coefficient ratios and the least formal term

The multiplicative theorem also determines the scale at which the terms of the divergent transseries stop decreasing. This is a statement about formal terms, not yet an analytic remainder theorem.

Proposition 11.1 (Ratio law throughout the superlinear range). *Under (2.1), for every fixed $p > 1$,*

$$\frac{u_n}{u_{n-1}} \sim \frac{1}{\tau_n} \sim e^{pr_n} \sim a \left(\frac{p}{p-1}\right)^p \frac{n^{p-1}}{(\log n)^p}, \quad (11.1)$$

where τ_n comes from any sufficient core and r_n solves (2.16).

Proof. Fix a sufficient core and extend its saddle smoothly to real n . The implicit-function theorem applies because $\Delta_n > 0$. Differentiating the two stationary equations gives the useful exact identities

$$\frac{dz_n}{dn} = \frac{C_n}{\Delta_n}, \quad \frac{d \log \tau_n}{dn} = -\frac{H_n}{\Delta_n}. \quad (11.2)$$

For completeness, at fixed j the inner equation gives $\partial_n \log t_j = 1/B_j$, while $\partial_j \log t_j = -C_j/B_j$ and $\partial_n f'_n(j) = C_j/B_j$. Differentiating $f'_n(z_n) = 0$ proves the first identity; substituting it into the total derivative of $\log t_{z_n}$ proves the second.

The scales already established imply $z'_n = O(z_n/n)$ and $\tau'_n/\tau_n = O(1/n)$. Differentiating the finite analytic expressions in (2.11) therefore gives $d \log \Delta_n/dn = O(1/n)$. There is no dangerous cancellation here, since $\Delta_n \asymp n^2/z_n^2$ and the differentiated terms are individually $O(\Delta_n/n)$. Likewise $d \log R(\tau_n)/dn = O(\tau_n/n)$. The stationary exponent satisfies the exact envelope identity

$$\frac{d}{dn}(az_n^p G(\tau_n) - (n - z_n) \log \tau_n) = -\log \tau_n.$$

Thus $d \log \mathcal{A}_{n,M}/dn = -\log \tau_n + O(1/n)$. Taking a difference over one unit and using $u_n \sim \mathcal{A}_{n,M}$ proves the first equivalence in (11.1). The full-core and bare radii have ratio tending to 1: both are asymptotic to the right side of (2.9). The last two equivalences follow from (2.16). $\square$

Corollary 11.2 (Least-term index). *In the pure model, let $N(q)$ be any integer minimizing $u_n q^n$ over $n \geq 1$, for $q > 0$. Such a minimum exists, and as $q \downarrow 0$,*

$$N(q) \sim a^{-1/(p-1)} q^{-1/(p-1)} \left(\frac{\log(1/q)}{p} \right)^{p/(p-1)}. \quad (11.3)$$

Equivalently, for $q = e^{-x}$, $N(e^{-x}) \sim a^{-1/(p-1)} (x/p)^{p/(p-1)} e^{x/(p-1)}$.

Proof. The coefficients are positive, and their n th roots tend to infinity by Lemma 4.1; hence $u_n q^n \rightarrow \infty$ for each fixed $q > 0$, so a minimum exists. Its index tends to infinity as $q \downarrow 0$, because for every fixed n the ratio qu_{n+1}/u_n then tends to zero.

Let $x = \log(1/q)$ and $r = x/p$. The real index at which the bare radius equals q is exactly

$$\nu(q) = a^{-1/(p-1)} r^{1/(p-1)} (1+r) q^{-1/(p-1)}. \quad (11.4)$$

The smooth comparison e^{pr_n} in (11.1) is increasing and regularly varying with index $p-1$. For any fixed $\delta > 0$, the ratios qu_n/u_{n-1} are therefore below 1 on all sufficiently large indices up to $(1-\delta)\nu(q)$, and above 1 from $(1+\delta)\nu(q)$ onward. This follows uniformly from the eventual relative bounds in (11.1) and monotonicity of the smooth comparison; the finitely many small indices also have ratios below 1 for small enough q . Every minimizing index lies between these two bounds. Letting $\delta \downarrow 0$ gives $N(q) \sim \nu(q)$, whose leading form is (11.3). $\square$

Remark 11.3 (A boundary that must not be crossed without another proof). The corollary locates the least coefficient term. It does not show that a sectorial analytic solution differs from an optimally truncated series by at most that term, or by any fixed multiple of it. Such a conclusion needs uniform remainder estimates for the actual function. Keeping these two statements separate is particularly important for the rapidly growing feedback kernel.

12 Sharp reversion in the superquadratic regime

12.1 A rapid-coefficient convolution lemma

We isolate the signed part of the argument. Positivity was essential in the forward cutoff proof, so that proof must not simply be reused after formal reversion.

Lemma 12.1 (Endpoint domination and reversion). *Let $A(q) = q + \sum_{n \geq 2} a_n q^n$, with $a_n > 0$. Suppose there is a smooth, eventually convex function F such that*

$$a_n \sim e^{F(n)}, \quad \frac{a_n}{a_{n-1}} \sim \mathcal{R}_n, \quad \mathcal{R}_n \sim c \frac{n^\rho}{(\log n)^\kappa}, \quad c > 0, \quad \rho > 1.$$

For $A^{(-1)}(q) = q + \sum_{n \geq 2} \alpha_n q^n$,

$$\alpha_n = -a_n + (n+1)a_2a_{n-1} + o(na_{n-1}). \quad (12.1)$$

Proof. Set $B(q) = A(q)/q - 1 = \sum_{m \geq 1} b_m q^m$, so $b_m = a_{m+1}$. We first show

$$[q^N]B(q)^2 \sim 2b_1b_{N-1}, \quad [q^N]B(q)^2 \leq Cb_{N-1} \quad (N \geq 2) \quad (12.2)$$

for a fixed constant C . The summands with indices 1 and $N-1$ give $2b_1b_{N-1}$. For each fixed $j \geq 2$,

$$\frac{b_j b_{N-j}}{b_{N-1}} = O(\mathcal{R}_N^{-(j-1)}) = o(1).$$

To bound the middle uniformly, choose a large fixed integer L beyond the convexity threshold of F . On $L \leq j \leq N-L$, the function $F(j+1) + F(N-j+1)$ is convex, so its maximum occurs at the endpoints. The comparability $a_n \asymp e^{F(n)}$ gives

$$\sum_{j=L}^{N-L} \frac{b_j b_{N-j}}{b_{N-1}} \leq C_L N \frac{a_{N-L+1}}{a_N} = O(N \mathcal{R}_N^{-(L-1)}) = o(1)$$

when L is sufficiently large. The finitely many endpoint terms between 2 and $L-1$ are already negligible. This proves the first assertion of (12.2); increasing C covers the finitely many small values of N and gives its second assertion.

Induction by convolution now gives, for $N \geq \ell \geq 1$,

$$[q^N]B(q)^\ell \leq C^{\ell-1} b_{N-\ell+1}. \quad (12.3)$$

Indeed, convolving the induction hypothesis with B reduces the next step to (12.2) with total index $N - \ell + 2$.

Lagrange inversion for $A^{(-1)}$ yields

$$\alpha_n = \frac{1}{n} \sum_{\ell=1}^{n-1} (-1)^\ell \binom{n+\ell-1}{\ell} [q^{n-1}]B(q)^\ell. \quad (12.4)$$

The term $\ell = 1$ is $-a_n$, while (12.2) makes the term $\ell = 2$ equal to $(n+1)a_2a_{n-1}(1+o(1))$. We bound all remaining terms absolutely, so no unproved cancellation estimate is used.

For $3 \leq \ell \leq n/2$, the ratio hypothesis, uniformly on indices between $n/2$ and n , gives

$$\frac{a_{n-\ell+1}}{a_n} \leq \left(\frac{C_1}{\mathcal{R}_n} \right)^{\ell-1}.$$

Also $n^{-1} \binom{n+\ell-1}{\ell} \leq 2(2n)^{\ell-1}/\ell!$. Thus the normalized absolute sum of these terms is $O((n/\mathcal{R}_n)^2)$. Because $n/\mathcal{R}_n \rightarrow 0$, this is $o(n/\mathcal{R}_n)$, or $o(na_{n-1}/a_n)$.

For $\ell > n/2$, the numerator coefficient in (12.3) is at most a constant times $a_{\lfloor n/2 \rfloor + 1}$. The ratio hypothesis implies

$$\log \frac{a_{\lfloor n/2 \rfloor + 1}}{a_n} = -\frac{\ell}{2} n \log n + O(n \log \log n).$$

This beats the $e^{O(n)}$ bound for the binomial factors and C^ℓ , and also the number of summands. Their contribution is smaller than $o(na_{n-1})$. Combining the three ranges proves (12.1). $\square$

12.2 Checking the hypotheses for the feedback coefficients

Proof of the reversion and ratio parts of Theorem 2.4. For $p > 2$, Proposition 10.1 gives $u_n \sim S_n$. Extend the positive solution r_n of (2.16) smoothly to real $x > 0$. Implicit differentiation gives

$$r'(x) = \frac{p-1}{x(p+1/r(x) + (p-1)/(1+r(x)))} \sim \frac{p-1}{px}. \quad (12.5)$$

Let $F(x) = \log S_x$. The derivative of the optimized bare exponent with respect to x is $-\log t = pr(x)$; the derivative in the optimizing action vanishes. The logarithmic determinant contributes only $O(1/(x \log x))$. Consequently

$$F'(x) = pr(x) + O\left(\frac{1}{x \log x}\right), \quad F''(x) \sim \frac{p-1}{x} > 0. \quad (12.6)$$

The second estimate follows either by differentiating the explicit r -formula once more or directly from (12.5); its absolute error is $O(1/(x \log x))$, hence its relative error is $O(1/\log x)$. Thus F is eventually convex.

The mean-value theorem and (12.6) show $S_n/S_{n-1} \sim e^{pr_n}$. The same is true for u_n/u_{n-1} because $u_n/S_n \rightarrow 1$. From (2.16),

$$e^{pr_n} = \frac{an^{p-1}}{r_n(1+r_n)^{p-1}} \sim a \left(\frac{p}{p-1}\right)^p \frac{n^{p-1}}{(\log n)^p}. \quad (12.7)$$

Lemma 12.1 applies with $\rho = p-1 > 1$ and $\kappa = p$, proving (2.20). In particular,

$$-\frac{v_n}{u_n} = 1 - \frac{(n+1)u_2}{e^{pr_n}} + o\left(\frac{n}{e^{pr_n}}\right), \quad (12.8)$$

and $v_n < 0$ for all sufficiently large n . The forward parts of the theorem were proved in Proposition 10.1 and its quadratic corollary. $\square$

12.3 The exact point where the proof stops

The small parameter in the inverse proof is

$$\epsilon_n := \frac{n}{\mathcal{R}_n} \asymp n^{2-p}(\log n)^p.$$

It tends to zero precisely in the superquadratic range considered here. At $p = 2$ it instead grows logarithmically, and the higher powers in (12.4) cannot be discarded by this argument. The forward cutoff theorem remains valid there, but its positivity is lost on reversion. We do not extend (2.20) into that range by analogy or by extrapolating numerical ratios.

Editorial note (ProveIt, 2026-09-29). The exponential (n th-root) scale of the inverse coefficients for $1 < p \leq 2$ is settled in limsup form by the later package [11], which takes two forward results of [2] as inputs: for $\lambda_j = aj^p$ its Theorem `thm:refinedfeedback` gives $\limsup_n (|v_n| \log(n+e))^{pn}/(n!)^{p-1})^{1/n} = a(p/(p-1))^p$, the forward value. For $p > 2$ this value already follows from (2.20). At $p = 2$ the factors separating the conjectured inverse equivalent from u_n are $e^{o(n)}$, so that result neither proves nor contradicts Conjecture 14.2; for the claimed proofs of the conjecture see the note after it.

13 Algorithms, exact checks, and numerical diagnostics

13.1 Positive coefficient recurrence

Let $E_{j,m} = [q^m]e^{\lambda_j \widehat{U}(q)}$ and $E_{j,0} = 1$. Differentiation of the exponential gives the triangular recurrence

$$u_n = \sum_{j=1}^n w_j E_{j,n-j}, \quad E_{j,m} = \frac{\lambda_j}{m} \sum_{k=1}^m k u_k E_{j,m-k} \quad (m \geq 1). \quad (13.1)$$

For nonnegative input, every term in these sums is nonnegative. Computing in logarithms with a log-sum-exp operation therefore avoids both overflow and subtractive cancellation. Only the triangle $j + m \leq N$ is needed through order N . The direct implementation takes $O(N^3)$ arithmetic operations and $O(N^2)$ storage; this is not a bit-complexity bound for large-integer arithmetic.

For integer slopes, factorial normalization avoids rational intermediate values. Put $b_n = n!u_n$ and $\mathcal{E}_{j,m} = m!E_{j,m}$. Then

$$b_n = \sum_{j=1}^n w_j \frac{n!}{(n-j)!} \mathcal{E}_{j,n-j}, \quad (13.2)$$

$$\mathcal{E}_{j,m} = \lambda_j \sum_{k=1}^m \binom{m-1}{k-1} b_k \mathcal{E}_{j,m-k}. \quad (13.3)$$

The supplied exact implementation uses $w_j = 1$, integer a and integer p . It computes the inverse in the same factorial normalization by the partial Bell-polynomial recurrence for the identity $\widehat{U}(\widehat{V}(q)) = q$.

13.2 Evaluating the finite-core formula

For numerical evaluation one need not first compute a long Taylor expansion of G_M . Solve simultaneously for (t, u, z) :

$$u = \Phi_M(t, u), \quad \log t + apz^{p-1}u = 0, \quad z + az^p t G'_M(t) = n. \quad (13.4)$$

The required derivatives are finite expressions:

$$G'_M = \frac{\Phi_t}{1 - \Phi_u}, \quad (13.5)$$

$$G''_M = \frac{\Phi_{tt} + 2\Phi_{tu}G'_M + \Phi_{uu}(G'_M)^2}{1 - \Phi_u}. \quad (13.6)$$

All partial derivatives are evaluated at (t, u) . The pure-model program uses logarithms of the three positive unknowns, initializes with the bare saddle, and checks the regular branch $\Phi_u < 1$, a positive determinant, and the numerical residual. These are floating-point checks, not interval certificates. The existence theorem ensures the intended small branch for sufficiently large n ; it does not supply an explicit universal starting order for a generic nonlinear root finder.

Each evaluation of the core and its derivatives uses only M summands. Thus the finite-core formula is useful even when storing the exact coefficient itself is impractical. A larger sufficient core can improve finite-order accuracy substantially without changing the limit theorem.

13.3 What was actually checked

The package records exact coefficients and formal inverse coefficients through degree 300 for $(a, p) = (1, 2)$ and $(1, 3)$. It contains:

- 36 independent comparisons of the positive recurrence with the partition formula: degrees 1 through 18 in the two models;
- 96 independent one-tail marking comparisons: degrees 1 through 16, cutoffs $M = 1, 2, 3$, in both models;
- 598 exact inverse-composition residual checks: degrees 2 through 300 in the two models.

All of these checks passed. The inverse coefficients are obtained from a recurrence enforcing the composition identity; the residual checks are not presented as an independent second proof of formal inversion. By contrast, the partition enumeration and the one-tail enumeration use different finite algorithms from the differential recurrences.

Separately, the log-domain positive recurrence was run through degree 1000 in both models. Comparing it with the exact coefficients through 300 gave maximum absolute discrepancies in $\log u_n$ of approximately 6.83×10^{-13} for $p = 2$ and 9.10×10^{-13} for $p = 3$. These agreement figures are diagnostics in ordinary floating-point arithmetic, not rigorous error bars.

13.4 The effect of the finite-core cutoff

Table 1 compares the computed coefficient u_{1000} with $\mathcal{A}_{1000,M}$ from (2.12), for $a = 1$. The quadratic $M = 1$ core is insufficient and has a large missing factor. The smallest sufficient core is not necessarily the most accurate choice at a moderate degree; adding a few more small actions improves the numerical approximation markedly.

Table 1: Finite-core diagnostics at $n = 1000$, $a = 1$. The ratio uses the log-domain recurrence; no interval certification is asserted.

p	M	$u_n/\mathcal{A}_{n,M}$	saddle action z_n
2	1	373.908262191	292.05833
2	2	1.051182297	290.22703
2	3	1.000393348	290.19821
2	4	0.999965103	290.19784
3	1	1.045788395	235.38637
3	2	0.999941539	235.37627
3	3	0.999938904	235.37627
3	4	0.999938904	235.37627

For the pure quadratic model at $n = 1000$, the ratio to the bare formula is approximately 124136.898, while the ratio to the explicit quadratic correction (2.19) is 1.19599044. The exact four-action core improves the ratio to approximately 0.99996510. The large bare error is not a finite-precision accident: its divergence is proved by the quadratic corollary. Conversely, the small observed four-core error is not a proven finite- n guarantee.

13.5 Inverse diagnostics and a deliberately separate conjecture

The middle column of Table 2 shows $-v_n/u_n$ from exact integer-normalized coefficients. The superquadratic ratios move toward the proved limit 1, albeit slowly. The quadratic ratios are far smaller. The last column compares with the conjectural extension discussed in Section 14; that column is exploratory, not a theorem check.

No finite collection of these comparisons proves the cutoff, the saddle limit, the central limit theorem, or the inverse asymptotic. Their role is to detect normalization errors, missing factors, and implausible finite-order behavior in formulas proved by other means.

Table 2: Exact-coefficient inverse diagnostics, $a = 1$. Here $\tilde{V}_n = -S_n \exp(-2k_n^0/(j_n^0)^{p-1})$ is an exploratory comparison; its quadratic asymptotic status is conjectural.

p	n	$-v_n/u_n$	$v_n/\tilde{V}_n$
2	50	0.000519898	0.649597
2	100	0.000121097	0.757155
2	200	0.000018894	0.835910
2	300	0.000005371	0.870969
3	50	0.571305779	0.978478
3	100	0.667665290	0.992327
3	200	0.751191069	0.997274
3	300	0.793480085	0.998512

14 Further research questions and a concrete inverse conjecture

14.1 Quadratic and subquadratic inverse asymptotics

The most immediate continuation is a finite-core theorem for the signed inverse coefficients when $1 < p \leq 2$. There is an exact identity suggesting a route, but the needed cancellation control has not been proved here.

Proposition 14.1 (Exact inverse response to one marked tail). *Let $Q = G_M^{(-1)}$. Fix $j > M$ and solve*

$$U_s(q) = \Phi_M(q, U_s(q)) + sq^j e^{aj^p U_s(q)}, \quad V_s = U_s^{(-1)}.$$

Then the formal derivative of the inverse at $s = 0$ is

$$\partial_s V_s(u)|_{s=0} = -Q'(u)R_M(Q(u))Q(u)^j e^{aj^p u}. \quad (14.1)$$

Proof. The direct response at $s = 0$ is $T_j(q) = R_M(q)q^j e^{aj^p G_M(q)}$ by Lemma 3.2. Differentiate $U_s(V_s(u)) = u$ at $s = 0$: $T_j(Q(u)) + G'_M(Q(u))\partial_s V_s(u)|_{s=0} = 0$. Use $Q' = 1/G'_M(Q)$ and $G_M(Q(u)) = u$. $\square$

For $M \geq 2$ in the pure model, $Q(u) = u - (a+1)u^2 + O(u^3)$. Thus the large power $Q(u)^j$ in (14.1) suggests an exponential penalty of leading form $\exp(-(a+1)ju)$, not the positive forward factor $\exp((a+1)ku)$. At the bare saddle this suggests the following candidate.

Conjecture 14.2 (Quadratic inverse equivalent). For the pure model with $p = 2$ and every fixed $a > 0$,

$$v_n \sim -S_n \exp(-(1 + 1/a)r_n). \quad (14.2)$$

The exact inverse-response identity does not prove that the one-tail response exhausts the inverse coefficient. That is the missing step. The quadratic data in Table 2 are consistent with the candidate, but do not establish its constant, sign for all large n , or its limit. For $p = 3$, convergence of the table's last-column ratio to 1 already follows from $v_n \sim -S_n$, because that column's extra exponential factor tends to 1; no refined rate follows from it. A useful attack would reorganize the signed Lagrange formula into an analytic core before bounding its remaining tail, rather than estimate individual inverse-composition powers absolutely as in the $p > 2$ proof.

Editorial note (ProveIt, 2026-09-29). Conjecture 14.2 remains stated here as this article's conjecture. Two later packages, filed in batch 49 (see docs/incoming/README.md), each claim a proof of it; neither proof has been independently reviewed, and neither package cites the other. They are [12], Theorem thm:main, and [13], Theorem thm:main. Both use r_n and S_n

of (2.16) and (2.17) and follow the route suggested above, resumming an analytic inverse core before bounding the signed tail: the first merges multiple large actions in a positive majorant, the second uses an exact quadratic merger defect. Their exact inverse coefficients agree at every common degree (through 220 for $a = 1$ and 180 for $a = 2$). Neither hypothesis set contains the other. The first allows finitely many nonnegative changes of weights and slopes and eventual slopes $aj^2 + dj + e$, with factor $\exp((d - \mu)r_n/a)$, $\mu = \lambda_1 + w_2$; the second allows arbitrary real changes of finitely many weights and slopes but an exact tail aj^2 , with factor $\exp(-(\lambda_1 + w_2)r_n/a)$. In the pure model both factors reduce to $\exp(-(1 + 1/a)r_n)$, which is (14.2). A third, `Signed_Condensation_Sharp_Quadratic_Reversion/` (batch 50), claims it under the first package's hypotheses (its Theorem `thm:main`) and adds the least formal inverse term; its exact coefficients agree with both (through degree 240 for $a = 1$ and 160 for $a = 2$). It has not been independently reviewed.

14.2 Regularly varying feedback and critical logarithmic boundaries

Replace the pure tail by $\lambda_j \sim aj^p(\log j)^\gamma$, with controlled derivatives or finite differences. The leading radius is then expected to have scale $n^{1-p}(\log n)^{p-\gamma}$. The corresponding omitted-action diagnostic would be

$$n\tau_n^M \asymp n^{1-M(p-1)}(\log n)^{M(p-\gamma)}.$$

At $M(p-1) = 1$, this predicts three different possibilities: a vanishing correction for $\gamma > p$, a finite nonzero correction for $\gamma = p$, and a divergent correction for $\gamma < p$. Establishing a theorem at that boundary requires a sharper replacement for the power-tail transfer bound, and potentially a Poisson correction when the scale tends to a constant. The power-tail theorem does not automatically cover these alternatives.

Editorial note (ProveIt, 2026-09-29). The later package [10] obtains such a Poisson correction at $M(p-1) = 1$ by letting the exponent approach the boundary, $p_n = 1 + 1/M + ((M+1) \log \log n + \theta)/(M \log n)$ (Theorem `thm:window`); the $(\log j)^\gamma$ tails proposed here are not treated there and remain open.

14.3 Joint laws for the finite background cloud

The present local theorem concerns the large action after conditioning on one tail. Determine the joint law of the numbers of actions $2, \dots, M$ and their dependence on J_n . Natural scales are $n\tau_n^{d-1}$, but their exact means and covariances depend on the core's implicit response. One can introduce finitely many marking variables in Φ_M and differentiate the saddle equations. The desired theorem would give Gaussian limits for diverging counts, Poisson limits for bounded counts, and a quantitative total-variation approximation where appropriate. It should explicitly distinguish this coefficient model from the fixed product-weight random-allocation models in [5, 6].

Editorial note (ProveIt, 2026-09-29). Two later packages answer this question in special cases. The package [10] treats the single count $N_{M+1,n}$ of first omitted actions near the cutoff boundary: Poisson limits in total variation for bounded intensity, Gaussian limits for diverging intensity, and independence from the largest action (Theorems `thm:probability` and `thm:joint`). The package [9] treats $p = 2$, $a = 1$: the number of actions equal to two is close in total variation to a Poisson law of mean r_n^2 , jointly with the largest action (Theorem `thm:probability`), and with high probability every action other than the largest is one or two. For other p , and for $p = 2$ with $a \neq 1$, the joint law of the counts of actions $2, \dots, M$ remains open.

14.4 Uniformity as the growth exponent approaches one

The minimum sufficient core diverges as $p \downarrow 1$. Moreover, when $(p - 1) \log n$ is bounded, the proposed giant size is no longer small relative to n . Both the fixed-core analytic estimates and the coarse localization need replacement. Develop a double-scaling theorem in which $p = p_n \downarrow 1$ and determine the transition between a macroscopic fraction of total action and the $n/\log n$ scale. A uniform result here would connect the convergent linear-feedback boundary with the divergent superlinear regime, rather than merely taking a limit in a nonuniform formula.

14.5 Quantified errors and certified saddle computation

The current asymptotic theorems have $o(1)$ relative errors without explicit universal onset values. Convert the mass-transfer inequality, the Cauchy arc bounds, and the action-saddle Taylor estimates into explicit constants. Then combine interval evaluation of the finite sums in (13.4) with a bracket for the unique coarse saddle. The target is a computable pair of bounds $L_n \leq u_n/\mathcal{A}_{n,M} \leq U_n$ with $L_n, U_n \rightarrow 1$ and a specified domain of validity. Floating-point residuals alone, even when small, do not provide that certificate.

14.6 Higher-order expansions and accuracy-dependent cores

A complete expansion should combine higher Cauchy-saddle terms, a discrete Edgeworth expansion for the action sum, and omitted-background corrections. For a target algebraic accuracy, the natural first estimate is to choose a core with $n\tau_n^M$ below that accuracy scale. This suggests stricter cutoffs than $M(p - 1) > 1$, but the logarithmic factors and the analytic-saddle errors must be tracked together. Determine the smallest core for a prescribed relative error $O(n^{-K})$, and whether the resulting expansion can be organized in a reusable symbolic coefficient calculus.

14.7 From least formal terms to actual analytic remainders

Corollary 11.2 determines a formal truncation scale for every $p > 1$. The earlier feedback article constructs left-sector analytic solutions [2], but a useful continuation would establish uniform remainder estimates strong enough to compare those solutions with truncation at that scale. Directional summability or acceleration requires further analytic growth information; neither follows from a coefficient ratio law alone. A precise target is a sectorial remainder bound with an explicit action-dependent exponent, followed by an analysis of the Stokes ambiguity between different realizations when such ambiguity exists.

14.8 A staged formalization in Lean

The shortest useful formalization path begins with finite algebra rather than the full saddle theorem. Formalize the positive Lagrange coefficient identity, its finite multiplicity form, and the marked one-tail response; then the convex mass-transfer inequality and the preimage-count bound. These are separate from the analytic implicit-function and contour estimates. The existing transseries support and asymptotic-scale library provides context, but no claim is made that it already supplies this entire analytic chain. A later stage can package shrinking-radius saddle estimates with uniform hypotheses, followed by the lattice Gaussian summation and signed inverse lemma. This decomposition would make a partially completed formalization useful without labeling the unformalized remainder as checked.

15 Conclusion

For power feedback, the passage from logarithmic estimates to relative asymptotics is controlled by a finite but sharply determined amount of background data. The correct reduction is not simply to one large action, but to one large action dressed by a sufficient analytic core. The exact threshold is $M(p-1) > 1$, and its strict boundary is forced by an explicit positive two-tail contribution.

Once this reduction is established, two ordinary saddle calculations give the full coefficient factor and a moving Gaussian law. They also expose the superpolynomial factor hidden in the quadratic logarithmic estimate and locate the least formal term. A separate rapid-coefficient argument handles formal reversion when $p > 2$. The quadratic inverse, quantitative analytic remainders, and uniform transition at $p = 1$ remain focused next problems, not conclusions silently assumed by the present results.

A Proof dependencies and scope audit

The proof order avoids circular use of the cutoff theorem. First the positive coefficient formula and coarse localization are proved. The upper side of the cutoff uses a combinatorial transfer. Independently, the saddle formula is proved for $w_{n,M}$ for every fixed core, including insufficient ones. That independent formula is then used to prove failure below the cutoff. Only after both sides are available is the unconditional fluctuation statement obtained.

Result	Main ingredients	Proven range
Positive coefficients and one-tail identity	Formal Lagrange inversion; analytic implicit core; marking	Assumptions (2.1)
Macroscopic localization	Convexity, Stirling bounds, one-dimensional strict maximum	Every fixed $p > 1$
Sufficient core	Slope transfer and explicit preimage count	$M(p-1) > 1$
One-tail coefficient equivalent	Uniform Cauchy saddle; moving lattice saddle	Every fixed $M \geq M_0$
Necessary core	Exact two-mark response and saddle multiplier	Failure for $M(p-1) \leq 1$
Gaussian laws	Saddle window and tail control	Local conditional; weak unconditional for sufficient M
First explicit correction	Analytic perturbation of optimized exponent	$p > 3/2$
Ratio and least-term index	Smooth saddle motion; regular variation	$p > 1$; least-term statement for pure model
Sharp signed reversion	Endpoint convolution; absolute high-power bound	Pure model, $p > 2$
Quadratic inverse equivalent	Exact response plus unproved tail dominance	Conjecture only

The ordinary analytic implicit-function theorem, Cauchy's coefficient formula, Stirling's estimate, elementary convexity, and the Gaussian integral are classical tools used in the proofs. Numerical agreement is not an additional axiom or a replacement for any of them. No theorem in this document has been compiled as Lean code. No repository file or branch was modified in preparing the package.

The main claims are stated for nonnegative weights and slopes with an *exact power tail after finitely many exceptions*. Signed primitive weights, an arbitrary regularly varying tail, a growing core, or a parameter p depending on n require additional arguments. In particular, a theorem about one of those larger classes is not implicit in our use of the word universality.

B Notation and normalization

Symbol	Meaning
q	Formal expansion variable; $q = e^{-x}$ gives the exponential transseries interpretation.
w_j, λ_j	Primitive weight and feedback slope of action j .
u_n, v_n	Ordinary coefficients of $\widehat{U}$ and its formal inverse. They are not exponential-generating-function coefficients.
G_M, R_M	Analytic finite core and its linear response near zero.
$W_M, w_{n,M}$	Formal series and coefficient for exactly one action above M .
K_n, E_n, J_n	Random number of parts, one plus excess action, and largest part in the normalized positive Lagrange measure.
z_n, τ_n	Real moving action saddle and its small coefficient radius.
$k_n = n - z_n$	Deterministic complement of the saddle action; not the random number of parts K_n .
B_n, C_n, H_n, Δ_n	Derivative quantities in (2.11); $\sigma_n^2 = B_n/\Delta_n$.
r_n, j_n^0, k_n^0, S_n	Bare saddle ratio, action, complement, and coefficient approximation.
$b_n = n!u_n$	Integer normalization used only in the exact-computation section and output files.
$\sim$	Ratio tends to one, except when explicitly applied to logarithms. All displayed asymptotics fix the model parameters first.

For the pure model the least sufficient core is $M_*(p)$ from (2.7). With arbitrary initial perturbations the theorem is applied with any $M \geq M_0$ also satisfying the strict cutoff. This ensures that every action subjected to the tail transfer has precisely the assumed power slope and unit weight.

C Reproduction and source provenance

The main TeX file is self-contained: its bibliography and numerical tables are embedded, with no external figures or bibliography database required. Compile it twice with `pdflatex`. The companion files are `README.md`, `SOURCES.md`, `PROOF_STATUS.md`, a `Makefile`, three Python programs, and their recorded data. A Python environment with NumPy and SciPy is needed only for numerical saddle diagnostics and table generation. The exact program uses the standard library alone.

```
python code/verify_exact.py --order 300
python code/diagnostics.py --order 1000 --powers 2 3
python code/report_tables.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The stored CSV files with integer-normalized coefficients are exact. The stored logarithms, nonlinear-solver residuals, and ratios to saddle formulas are floating-point data. Re-running the scripts may change last printed digits or timings across platforms. The article's asymptotic proofs do not depend on those digits or on any external computation service.

The repository motivation is tied to the pinned source identified in Section 1. The literature check was targeted, not an exhaustive priority search. In particular, the paper treats the existence of related condensation and rapid-series methods as prior work, while proposing the exact finite-core reduction and the associated model-specific asymptotics as the contribution to be independently reviewed.

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